

Create a prompt document for my rsna project competition
Key overview captured:

- RSNA Knee Abnormality Detection competition.
- Goal: detect **12 clinically important knee abnormalities** from MRI.
- Dataset includes **MRI + original radiology report** → multimodal learning.
- Evaluation: **macro-averaged ROC-AUC across 12 targets**.
- Submission requires confidence scores for all 12 labels per **StudyInstanceUID**.
- Kaggle notebook constraints: **≤9h CPU/GPU runtime, no internet, external public data/pretrained models allowed, submission.csv required**.
- There is also an **efficiency track** considering predictive performance + runtime.

Dataset — Antigravity-ready

- **Task:** Predict per-study probability for **12 knee MRI abnormalities**.
- **Data:** Knee MRI studies containing multiple DICOM series from a single scanning session.
- **Coverage:** Diverse international imaging sites, scanners, protocols, and populations.
- **Labels:** Only a small subset of training studies has direct per-condition labels; radiology reports are provided to derive labels for remaining studies.
- **Reports:** Original free-text radiology reports; may be in multiple languages.

train.csv

- One row per training study.
- **StudyInstanceUID:** unique study ID; matches **train_series/<StudyInstanceUID>/.**
- **Report:** free-text radiology report.
- 12 binary labels:
 - **ACL** — anterior cruciate ligament injury
 - **MCL** — medial collateral ligament injury
 - **Medial Meniscus** — medial meniscus tear
 - **Lateral Meniscus** — lateral meniscus tear
 - **Medial OA** — medial tibiofemoral osteoarthritis
 - **Lateral OA** — lateral tibiofemoral osteoarthritis
 - **PF OA** — patellofemoral osteoarthritis
 - **Effusion** — joint effusion / excess fluid
 - **Synovitis** — joint-lining inflammation
 - **Baker's** — Baker's cyst

- **Contusion** — bone contusion / bone bruise
- **Fracture** — fracture

train_series.csv

- One row per MRI series/acquisition; each study contains multiple series.
- **StudyInstanceUID**: parent study ID.
- **SeriesInstanceUID**: unique series ID; matches `train_series/<StudyInstanceUID>/<SeriesInstanceUID>/`.
- **Fluid_Sensitive**: 1 if fluid-sensitive sequence (T2, PD, STIR, etc.), else 0.
- **Fat_Suppression**: 1 if fat suppression is applied, else 0.
- **Anatomical_Plane**: **Sagittal**, **Coronal**, or **Axial**.
- **Fluid_Sensitive** and **Fat_Suppression** are often correlated but **not equivalent** in every case.

DICOM structure

- `train_series/<StudyInstanceUID>/<SeriesInstanceUID>/<SOPInstanceUID>.dcm`
- Each `.dcm` = one MRI slice.
- Typical series: **20–45 slices**, median ≈ 30 , with some series extending to hundreds.
- Intensities, orientations, and resolutions vary across studies/series.
- Transfer syntaxes include:
 - Uncompressed Explicit VR Little Endian
 - JPEG Lossless
 - JPEG 2000
 - Implicit VR Little Endian
- DICOMs contain only an allowlisted set of **86 metadata tags**.

Test data

- **test.csv**: study IDs; $\approx 1,300$ **test studies**.
- **Report** is **not provided during testing**.
- **test_series.csv**: same schema as **train_series.csv**.
- **test_series/**: same DICOM structure as training data.
- Example test files are replaced by actual test data during scoring.

Submission

- **sample_submission.csv**: valid submission with all target probabilities set to **0.5**.
- Required predictions: per-study probabilities for all **12 labels**.

Dataset scale

- **819,640 files**
- **569.76 GB**
- Types: `.dcm`, `.csv`
- Main components: `train_series/`, `test_series/`, `train.csv`, `train_series.csv`, `test.csv`, `test_series.csv`, `sample_submission.csv`.
- License: **Subject to Competition Rules**.

Important dataset constraint

- Abnormality prevalence is **not guaranteed to be the same** across training, public leaderboard, and final evaluation datasets.

Competition Rules — Antigravity-ready

- **Entry:** Accepting the official rules is required to participate/submit; only **one Kaggle account** may be used.
- **Sponsor:** Radiological Society of North America (RSNA).
- **Maximum team size: 5 members.**
- **Submissions:** Maximum **5/day**; up to **2 final submissions** for judging.
- **Team mergers:** Allowed subject to team-size and submission-count limits and the merger deadline.

Data usage & security

- Competition Data may be used for competition participation, Kaggle forums, academic research, and education under the specified license/terms.
- Do **not** transmit, duplicate, publish, redistribute, or provide Competition Data to anyone not participating in the competition.
- Report unauthorized access/transmission to Kaggle.

External data / models / tools

- External data may be used if it is **publicly available and equally accessible at no cost**, or meets the competition's Reasonableness criteria.
- External models/tools are allowed unless specifically prohibited, but must be reasonably accessible/minimal cost.
- AMLT/AutoML tools are permitted if appropriately licensed.

Code sharing

- No **private code sharing outside the official team** during the competition.
- Public competition code may be shared through the relevant Kaggle competition forum/notebooks.

- Open-source code used for submissions must comply with the specified OSI-approved licensing requirements.

Submission restrictions

- Do not use **hand labeling or human prediction of validation/test records**.
- Submissions must follow the competition website's required format and deadlines.
- Only one unique Kaggle account may be used for submissions.

Winner requirements

- Winners must provide:
 - Final model software/code
 - Training code
 - Inference code
 - Model weights
 - Documentation
 - Required computational environment/dependencies
 - Instructions sufficient to reproduce the submission
- Model weights must be provided as a **public Kaggle dataset** linked to inference/submission code.
- Detailed methodology may be required, including architecture, preprocessing, loss function, training details, hyperparameters, etc.

Licensing

- Winner submission/source code: **CC-BY-NC 4.0** under the competition-specific winner license.
- Commercially available third-party software does not necessarily need to be relicensed; incompatible-license input data/pretrained models may remain under their original licenses.

Eligibility

- Must have a registered Kaggle account.
- Must meet the applicable age requirement.
- Certain sanctioned/restricted jurisdictions and persons/entities under applicable U.S. export controls are excluded.
- Competition is otherwise open worldwide subject to local laws and prize restrictions.

Leaderboard

- **Public leaderboard:** representative public test data.
- **Private leaderboard:** private test data and determines final standing.
- Private leaderboard results are used for final winner determination.

Core implementation implications

- Keep all Competition Data secure.
- Do not privately share project code/data outside the team.
- Ensure every external dataset, pretrained model, library, and tool satisfies competition accessibility/licensing rules.
- Build the pipeline so the **final model, weights, training/inference code, dependencies, and methodology can be reproduced** if required.

ALSO I CAN ONLY SUBMIT 5 SUBMISSION ONCE A DAY SO YOU WOULD HAVE TO COMPLETE THE PROJECT IN SMALL PARTS WHICH I CAN LATER ON CREATE NOTEBOOK ON KAGGLE PASTE OR UPLOAD MY WORK THERE AND THEN SUBMIT IT

THIS THE LEADERBOARD CHART ON KAGGLE
I NEED MY PROJECT ACCURACY EAXCTLY
LIKE TOP ONE SO WORK THAT WAY

1	Wassim Dobbi	 0.959	73	4h
2	rhoskeri	 0.959	74	1d
3	Scott Willis	 0.958	165	3h
4	TomKroumov	 0.958	78	12h
5	takoi & charmq	 0.957	169	12h

6

Parag & Ian

 0.957

131

12h

7

Volkopat

 0.957

124

6h

8

dalab

 0.956

143

14h

9

🏹 YOKKOISHO 🏹

 0.956

219

2h

10

Ex-Aggr

 0.956

177

12h

11

Brandon Low

 0.956

105

5h